

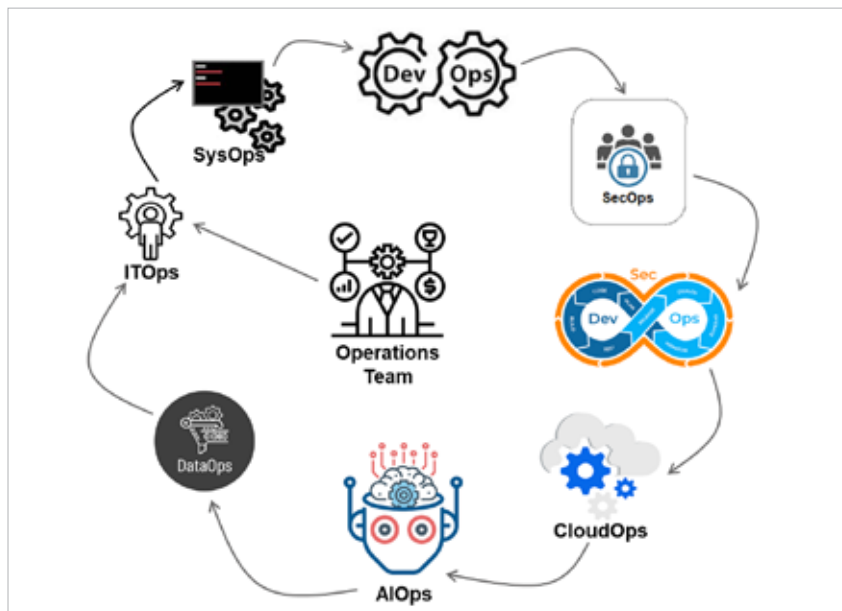


Figure 3: Relation and evolution of Ops activities

Advantages of Jenkins:

- Open source tool, easy to use and configure, and no need for any extra component to use it
- Supports Windows, Linux, UNIX, MacOS X, and others
- Automates all integration works, saving money and time spent on a project life cycle
- Easy to extend and modify; can test, build, automate and deploy code on different platforms.
- Detects and fixes issues, and has a diverse range of plugins
- Supports developers for early error detection

Official website: www.jenkins.io

Latest version: 2.281

CloudOps: The technical view

The IT operations landscape and life cycle stages are today more agile, cost-effective, smooth and well-defined, due to technologies like ITOps, CloudOps, DevOps, NoOps, AIOps, BizDevOps, DevSecOps, SysOps and DataOps.

These Ops have not grown to the next stage in a sequential manner but can be logically arranged, as

shown in Figure 3.

ITOps comprises traditional IT operations for IT management, networks, infrastructure, application management and technical support or help desk operations.

SysOps is more of infrastructure activities (or traditional software configuration management) to handle build, deployment, migration and managing systems in cloud platforms; it is the early stage of CloudOps.

DevOps creates agile teams for Development and Operations that work together in frequent iterations. The improved version of DevOps 2.0 is also called BizDevOps; it enables the business team (e.g., business analyst, consultant) to collaborate with the

DevOps team for product development.

SecOps focuses on the relationship between the security and risk management team and the operations team to strengthen various security activities like infrastructure security, network security, data security, application security, etc.

DevSecOps combines DevOps and SecOps for smooth interaction between the development, security and operations teams to design, develop, implement, build, deploy and maintain activities.

AIOps was originally defined by Gartner to handle Big Data platforms, machine learning platforms and analytics platforms. It also aims at automation to reduce the complexity of IT operations.

DataOps is a sliced down version of AIOps to handle data platforms. It includes data management, migration, validation and analytics, and sets up processes and standards for data management activities.

NoOps implies automating everything and spending nothing on the operations team. This is the ultimate aim of organisations. For example, Netflix and Facebook are focusing on development and not setting up an operations team for support.

CloudOps is based on four factors, as shown in Figure 4.

Cost: The monitoring and cloud analytics solution must be cost optimised for better cloud governance.

Consumption: Helps to derive template based options for CloudOps solutions like Terraform templates or CloudFormation/ARM templates in order to reduce the effort and complexity of setting up these solutions.

Observability: Though it is built on traditional services like monitoring, log handling, tracing and auditing, this is one of the key pillars of CloudOps. It ensures that reliability and agility in cloud governance can be handled easily.

Compliance: Security, risk management and compliance handling are key activities, and require swift action to ensure the solution blueprint

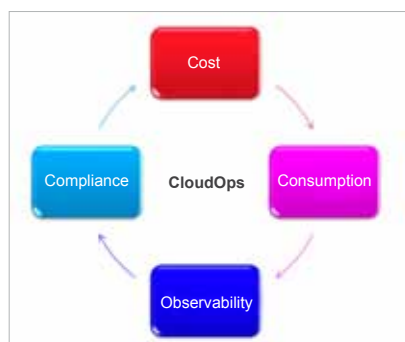


Figure 4: Four factors of CloudOps